SAI VAMSI KRISHNA RAMAVARAPU

DATA ANALYST

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SUMMARY

- Over 3+ years of experience in data analysis, data warehousing, and business intelligence, with a strong background in SQL and Python.
- Expert in SQL, including advanced queries, performance tuning, and optimization. Proficient in writing complex SQL queries, stored procedures, and functions to extract and manipulate data.
- Proficient in Python with extensive use of libraries such as NumPy, Pandas, and Scikit-learn for data analysis, statistical modeling, and machine learning. Skilled in data cleaning, transformation, and visualization.
- Experienced in data warehousing, including the design and implementation of OLTP and OLAP systems. Strong understanding of dimensional modeling, data integration, and ETL processes.
- Skilled in creating impactful visualizations with tools like Tableau and Power BI, enabling data-driven decision-making and business insights.
- Deep knowledge of data modeling principles, including star and snowflake schemas, fact and dimension tables, and relational data modeling.
- Strong experience with performance tuning and optimization of databases, including indexing, query optimization, and cache management.
- Adept at using data analysis tools and methodologies to support strategic decisions, including statistical analysis, predictive modeling, and trend identification.
- Proven ability to work with business users and senior management, translating complex data into actionable insights and facilitating requirements gathering sessions.

TECHNICAL SKILLS

Programming Language:	Python, SQL.
Databases:	MySQL, SQL Server
Libraries:	NumPy, Pandas, Matplotlib, SciPy, ggplot2, Scikit-learn, Seaborn, TensorFlow
Data Visualization Skills:	Tableau, Power BI, MS Excel, Alteryx.
Cloud Technology:	AWS (EC2, S3, Lambda, RDS, RedShift, Cloud Watch).
Methodologies:	SDLC, Agile/Scrum, Waterfall
Version Control Skills:	Git, GitHub.
Analytical Skills:	Statistical Modelling, Time Series Analysis, Predictive Analytics, Data Mining, Data Cleansing, Data Wrangling, Data Warehousing, Text Mining, Critical Thinking, Problem-solving, Communication, Business Analysis, Cost/Benefit Analysis, Impact Analysis, GAP Analysis, Risk Analysis, SWOT Analysis

EDUCATION

Master of Science in Business Analytics,
The University of Texas at Dallas — **Aug 2022 - May 2024**

Graduate Certificate in Applied Machine Learning, The University of Texas at Dallas — **May 2024**

Bachelor of Technology in Electronics and Communications,
Gandhi Institute of Technology and Management College — **Jun 2016 - Jun 2020**

PROFESSIONAL EXPERIENCE

Blue Cross Blue Shield, TX
Data Analyst — **Jul 2024 - Current**

- Led the analysis of complex datasets to identify trends, patterns, and actionable insights, driving strategic business decisions and contributing to a 15% increase in operational efficiency.
- Designed and developed interactive dashboards and reports using Power BI improving data visualization and accessibility for over 100 stakeholders across various departments.
- Conducted advanced statistical analysis and predictive modeling using Python and R, supporting initiatives that resulted in a 10% increase in sales revenue.
- Collaborated with cross-functional teams to gather data requirements and deliver tailored analytical solutions, enhancing data-driven decision-making processes.
- Ensured data quality and integrity through rigorous data cleaning and validation processes, reducing data inconsistencies by 20%.
- Presented findings and recommendations to senior management, aiding in the formulation of strategic initiatives and business plans.

Tata Consultancy Services, Bangalore, India
ETL Developer/ Assistant Systems Engineer — **Nov 2020 – May 2022**

- Developed and implemented Informatica ETL processes, streamlining data extraction from diverse sources and transforming it to meet business requirements. Delivered a 20% increase in data processing efficiency across three distinct data projects.
- Collaborated with database administrators to design and implement optimized SQL scripts for data loading, implementing efficient filtering mechanisms and reducing overall execution time of Informatica workflows by 30%.

- Conducted in-depth analysis of business processes, working closely with stakeholders to identify data integration needs, resulting in the creation of tailored Informatica ETL solutions aligned with strategic goals which improved data accuracy by 25% and reduced processing errors by 15%.
- Prepared technical documentation, including design specifications, data flow diagrams, and process documentation, to facilitate knowledge transfer resulting in a 40% reduction in onboarding time for new team members.

Beyond Key Solutions, India
Data Analyst Intern **Nov 2019 – Oct 2020**

- Assisted in the development and refinement of complex data models and interactive dashboards to support various client projects, significantly enhancing the client's ability to make informed, data-driven decisions and improving overall project outcomes.
- Conducted thorough data cleaning and preprocessing on large datasets, implementing advanced techniques to ensure the accuracy and reliability of data, which resulted in a measurable 25% improvement in data quality and consistency across multiple projects.
- Collaborated closely with senior analysts to perform in-depth statistical analyses, employing various statistical tools and methods to uncover actionable insights, ultimately contributing to a 15% increase in client satisfaction through more informed decision-making.
- Utilized SQL to efficiently query and manage extensive datasets, optimizing data retrieval processes and reducing query execution times by 30%, thereby improving the speed and efficiency of data analysis workflows.
- Created detailed and visually compelling reports and dashboards using Microsoft Excel and Power BI, translating complex data trends into easily understandable visualizations that helped clients make strategic business decisions with greater confidence.
- Actively participated in client meetings, presenting data-driven findings and strategic recommendations, which contributed to the development and refinement of long-term business plans, ensuring alignment with client goals and objectives.
- Leveraged AWS services, including EC2 for scalable computing and Lambda for serverless processing, to automate data workflows and enhance the efficiency of client projects, resulting in faster data processing times and reduced operational costs.

Siemens, Bangalore, India
Data Analyst Intern **May 2019 – Oct 2019**

- Worked on the PCGE software to design railway systems to incorporate electronic interlocking signaling system.
- Drafted blueprints for interlocking systems for six different railway stations and implemented them.
- Worked in the testing department to carry out various functionality tests on the hardware equipment to achieve 100% accuracy.
- Utilized SQL to query and manage large datasets, optimizing data retrieval and ensuring accurate information was available for designing and implementing railway interlocking systems.
- Developed and maintained detailed reports and dashboards in Power BI and MS Excel, providing critical insights into system performance and helping stakeholders make informed decisions throughout the project lifecycle.
- Managed version control using Git, ensuring seamless collaboration with team members by tracking changes, resolving conflicts, and maintaining a clear history of all project documentation and code related to the railway systems design and testing.

PROJECTS

EY Data Challenge - Machine Learning | Python

- Developed a K-Nearest Neighbors (KNN) classification model for the 2024 EY Open Science Data Challenge, demonstrating robustness and ability to handle non-linear decision boundaries, achieving an accuracy of 97%.
- Employed advanced API connections to access Microsoft Planetary Computer datasets, enabling efficient utilization of Sentinel-2, Landsat, and Sentinel-1 data for model training and validation.
- Incorporated polynomial feature transformation in the KNN model, coupled with stratified K-fold cross-validation and grid search, to enhance predictive accuracy and ensure optimal generalization across diverse input scenarios.

Facial Emotion Detection - NLP | Python

- Developed a Facial Emotion Recognition system using deep learning techniques, specifically implementing DenseNet169 with TensorFlow and Keras.
- Implemented a robust evaluation framework, assessing model performance using key metrics such as accuracy, loss, confusion matrix, and ROC-AUC scores across seven emotion categories.

Trucks Risk Assessment - Hadoop | PowerBI | Hive

- Applied big data technologies to analyze trucking movement, incorporating geographic data, vehicle details, average mileage, gas consumption, and other relevant information.
- Generated visual reports using PowerBI to highlight drivers with risk factors equal to or greater than 7.0, facilitating informed decision-making for risk mitigation.

Student Performance Prediction Model - Machine Learning | Python

- Developed a student performance prediction model to predict the final grade using various machine learning algorithms, including Linear Regression, KNN Regression, Support Vector Regression, and Ridge Regression.
- Collaborated with team members to analyze the dataset, identify relevant features, and make informed decisions for feature selection and model evaluation based on cross-validation techniques.

CERTIFICATIONS

Google Advanced Data Analytics - Link
Alteryx Foundation